EXPANDED DRAFT OF AN APPENDIX FOR *AN ORDER ON THE LEECH LATTICE FROM A $\mathbb{Z}_3$-SYMMETRIC TRIPLE-OCTONION PRODUCT*
HISTORICAL NOTE: INTEGRAL CAYLEY NUMBERS, 1923–1946

CLAUDE OPUS 4.7

Abstract. This is an expanded draft of the historical appendix. It is written *solely from the primary sources*: the four papers Dickson (1923), Kirmse (1924), Mahler (1942), and Coxeter (1946), each obtained and re-derived in exact arithmetic in this project (the scripts `verify_dickson_1923.py`, `verify_kirmse_1924.py`, `verify_kirmse_1924_forensic.py`, `verify_mahler_1942.py` and `verify_coxeter_1946.py` in `python_project/src/`, with companion notes in `paper/reviews/`); Mahler's covering-radius theorem is checked by sampling in exact arithmetic. No secondary source is taken on trust. A short form of this appendix, targeted at the main paper's length budget, is in `paper/historical_appendix.tex`.

Setting

The integral-Cayley-number construction (a maximal order in the real octonion algebra, supported on the E_8 lattice) was developed in four papers between 1923 and 1946. Each contributes one piece. Together they fix three facts that underlie the present construction: there are *seven* maximal orders containing the eight basis units; each is obtained from Kirmse's lattice J_1 (which is not an order) by a transposition of two imaginary basis units (a linear involution of $\mathbb{R}^8$); and the underlying lattice is E_8. Dickson exhibited one such order but counted three; the count of seven is Coxeter's, the second fact is the Coxeter–Bruck remedy of Kirmse, and Coxeter identified the lattice as E_8 (the vertex set of the honeycomb 5_{21}).

Dickson 1923

L. E. Dickson [Dickson1923], in Section 19 of *A new simple theory of hypercomplex integers*, gave the first construction. He built the octonions by Cayley–Dickson doubling of the quaternions, $x = q + Qe$ with q, Q quaternions, and the product (his equation (24))

$$(q + Qe)(r + Re) \;=\; (qr - \bar{R}Q) + (Rq + Q\bar{r})\, e,$$

and then determined the maximal sets of integers closed under his three axioms: closure C, contains unit U, integral norm N (his Section 6). From the norm form he derived that integer coordinates are either integers or half-integers, ran a case analysis, and exhibited (his (29)) the basis

$$i,\ j,\ k,\ p,\ e,\ W,\ Z,\ V, \qquad p = \tfrac{1}{2}(1 + i + j + k),$$

of an order, his *system (30)* (here denoted O_D). All 64 products of the basis (29) lie in O_D; the system is genuinely closed [Dickson1923].

Dickson then stated as **Theorem XV**:

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Drafted by the AI agent Claude Opus 4.7 (Anthropic) at the direction of Jens Köplinger, as part of the systematic investigation recorded in this repository (`prompt_logs/`). The version that appears in the main paper, as Appendix B, is owned by the principal investigator; this expanded form is a standalone reading copy and is intentionally longer than Appendix B..

> *The only maximal systems of integers of Cayley's algebra having properties C, U, N are (30) and the two systems obtained from it by cyclic permutations of i, j, k.*

This undercounts. Re-derived in this project, the true number of maximal orders for Dickson's product is *seven*, not three. The cause is in Dickson's own derivation: on p. 321, having reached a case in which the order contains the Hurwitz quaternions $\frac{1}{2}(\pm 1 \pm i \pm j \pm k)$, he writes

> *"... and hence all of Hurwitz's integral quaternions. We shall assume that the latter occur in our set in all cases."*

The assumption that every maximal order contains the Hurwitz quaternions is stronger than C, U, N. Of the seven maximal orders, *three* contain those quaternions, and those three are precisely O_D and its two $(i\,j\,k)$-cyclic images, exactly Dickson's Theorem XV triple. The four he missed are exactly the maximal orders that do not contain the integral quaternions.

So Dickson's *construction* is correct; his *count* is too small by four, by a qualifier that he stated openly in the derivation but that was absorbed into the theorem.

Kirmse 1924

J. Kirmse [Kirmse1924], working in Germany the following year, took up the problem with a different presentation. He fixed an octonion multiplication on the basis $\{1, i_1, \ldots, i_7\}$ by seven cyclic Fano triples (his table (1) on p. 64) and sought the maximal sets of octonion integers closed under multiplication. He derived a correct *necessary* form for the maximal orders (thirty E_8-type lattices satisfy it) but did not prove sufficiency. He then stated that there are *eight* such sets (*"Man findet acht solche ...,"* p. 70); in our reconstruction these are the eight of the thirty that are stable under multiplication by units, an unstated additional narrowing whose sufficiency Kirmse did not prove either. He described one of his eight in detail, his system J_1 (p. 70):

$$J_1 \;=\; \big[\, J_0,\; a_1,\; a_2,\; a_3,\; a_4 \,\big],$$

where $J_0 = \mathbb{Z}\langle 1, i_1, \ldots, i_7\rangle$ and

$$a_1 = \tfrac{1}{2}(1 + i_1 + i_2 + i_3), \qquad a_2 = \tfrac{1}{2}(i_4 + i_5 + i_6 + i_7),$$
$$a_3 = \tfrac{1}{2}(1 + i_1 + i_4 + i_5), \qquad a_4 = \tfrac{1}{2}(1 + i_2 + i_4 + i_7).$$

Kirmse's multiplication table is itself a correct octonion algebra: it is norm-multiplicative and satisfies all the Cayley identities. His error is in the lattice J_1: it is *not closed* under his own multiplication. The witness, computed in this project and matching Coxeter (1946) verbatim, is

$$a_1 \cdot a_3 \;=\; \tfrac{1}{2}(1 + i_1 + i_2 + i_3) \cdot \tfrac{1}{2}(1 + i_1 + i_4 + i_5) \;=\; \tfrac{1}{2}(i_1 + i_2 + i_4 + i_7),$$

and the product is not among the 240 elements of J_1 of unit norm (Kirmse p. 76). An enumeration of the 30 candidate E_8-type lattices under Kirmse's table yields exactly *seven* closed ones; J_1 is the spurious entry on Kirmse's list of eight.

So Kirmse's *table* is correct; his *exhibited order* is the defective one, and his *count* is too large by one.

Mahler 1942

K. Mahler [Mahler1942], eighteen years later, worked over Dickson's maximal order O_D (integral Cayley numbers "defined according to Dickson") and developed its arithmetic. His principal results are two.

- **Euclidean property** (Theorem 1, p. 126): *for every Cayley number X there is an integral Cayley number G with $N(X-G) \le \frac{15}{16}$*, sharper than Dickson's earlier $5/4$. On p. 127 Mahler states that the best constant is $\frac{1}{2}$, calls this "not easy to prove" and omits the proof.

- **Principal ideals**: *every left (or right) ideal in the ring of integral Cayley numbers is principal, generated by a multiple (in the ordinary sense) of an integral Cayley number of norm* 1, 2, *or* 4.

Re-derived in this project, Mahler's results are correct as stated; the printed paper contains one transcription error in equation (10), without consequence for the theorems. Mahler cites Dickson (1923) but not Kirmse; that citation chain is how the priority of Dickson's construction reaches Coxeter in the following paper.

Coxeter 1946

H. S. M. Coxeter [Coxeter1946], in *Integral Cayley numbers*, identifies Kirmse's error and supplies a corrected order. His Section 4 ("Kirmse's mistake") reproduces Kirmse's table as his (4.1) (the two tables are identical, verified) and exhibits the same nonclosure witness as above:

$$\tfrac{1}{2}(1+i_1+i_2+i_3)\cdot\tfrac{1}{2}(1+i_1+i_4+i_5) \;=\; \tfrac{1}{2}(i_1+i_2+i_4+i_7),$$

which, Coxeter notes, "is not one of the 240 units." Of the count he writes: *"Actually there are only seven, which presumably are the remaining seven of his eight"* (p. 566).

The remedy is due to **R. H. Bruck**. Coxeter writes: *"When I pointed this out to Bruck, he sent me a revised description of such a domain. Bruck's domain J can be derived from J_1 by transposing two of the i's."* That is, the correction is a transposition of two imaginary basis units of the octonions, a linear involution of $\mathbb{R}^8$. Any of the $\binom{7}{2} = 21$ such transpositions repairs Kirmse's J_1. Coxeter further notes that the 21 "fall into 7 sets of 3, each set having the same effect," yielding the seven maximal orders; this combinatorial structure is verified in this project (each set of 3 is three disjoint transpositions fixing one common imaginary unit, the "special unit" of the corresponding order).

Coxeter then develops the geometry of the corrected order. Its 240 norm-1 units are the vertices of Gosset's polytope 4_{21}; the order itself is the vertex set of the eight-dimensional honeycomb 5_{21} (his Sections 6–13). He proves the corrected order maximal in his Section 11.

A postscript (Section 14) records that Coxeter learned of Dickson and Mahler only after the manuscript was written, when Olga Taussky Todd drew his attention to Mahler, which in turn cites Dickson. Coxeter there identifies the cause of Dickson's Theorem XV undercount independently:

> *"... he asserted [9; 324] that there are only three systems which include all the eight basic units ... He missed the remaining four of the seven systems by adding the very artificial requirement [9; 321] that the integral Cayley numbers should include the integral quaternions* $\frac{1}{2}(\pm 1 \pm i \pm j \pm k)$*."*

Coxeter also observes that the bound with the best constant $\frac{1}{2}$, which Mahler calls "not easy to prove", is an immediate consequence of Section 11 of the 1946 paper. By 1946 the construction is complete.

Synthesis

The four papers together fix the construction the present paper rests on:

- **1923 (Dickson).** An integral-Cayley-number maximal order exists; the count, in the form Dickson's theorem gives, is three (under his stated extra assumption).
- **1924 (Kirmse).** An independent treatment; correct multiplication; an enumerated count of eight that includes one nonclosed candidate.
- **1942 (Mahler).** Arithmetic of the order: Euclidean with $N(X-G) \le \frac{15}{16}$ (best constant $\frac{1}{2}$, stated without proof); principal ideals.
- **1946 (Coxeter, with Bruck).** Kirmse's J_1 is not closed (an explicit one-line witness); Bruck's remedy is a transposition of two imaginary basis units; the 21 such transpositions

yield exactly the seven maximal orders; the lattice is E_8 (the vertex set of 5_{21}); Coxeter's postscript supplies the missing connection to Dickson and Mahler.

The count converges over the four papers ($3 \to 8 \to 7$), and the correction that turns Kirmse's J_1 into a genuine order is the *Coxeter–Bruck transposition* of two imaginary coordinate axes: a linear involution of $\mathbb{R}^8$.

It is a curious coincidence that this transposition would come in useful at the Leech-lattice level: the same linear involution of $\mathbb{R}^8$, but acting on a different object. The Coxeter–Bruck transposition is applied to a maximal-order candidate that sits asymmetrically in $\mathbb{R}^8$, and so changes which E_8 placement is considered. The present σ (Definition 3.1 of the main paper) is applied where the order is $L = D_8^+$, the coordinate-symmetric placement; hence $\sigma(L) = L$ and the E_8-level closure is preserved automatically ($L \cdot_\sigma L = \sigma(L \cdot L) \subseteq L$). The two transpositions are the same kind of map at different strata: at the Leech-lattice level rather than the maximal-order level, redirecting Wilson's sublattice conditions rather than repairing the choice of order.

References

[Dickson1923] L. E. Dickson, *A new simple theory of hypercomplex integers*, J. Math. Pures Appl. (9) **2** (1923), 281–326.

[Kirmse1924] J. Kirmse, *Über die Darstellbarkeit natürlicher ganzer Zahlen als Summen von acht Quadraten und über ein mit diesem Problem zusammenhängendes nichtkommutatives und nichtassoziatives Zahlensystem*, Ber. Verh. Sächs. Akad. Wiss. Leipzig, Math.-Phys. Kl. **76** (1924), 63–82.

[Mahler1942] K. Mahler, *On ideals in the Cayley–Dickson algebra*, Proc. Roy. Irish Acad. Sect. A **48** (1942), 123–133.

[Coxeter1946] H. S. M. Coxeter, *Integral Cayley numbers*, Duke Math. J. **13** (1946), 561–578.

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